

PRACTICAL WEEK – 3

NAME:CH.SAI VIHAMSH

ROLL:2520090090

Session-3

Task1: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
saivihamsh@Varshini-PC:~$ nano pweek3.c
saivihamsh@Varshini-PC:~$ gcc pweek3.c -o pweek3
saivihamsh@Varshini-PC:~$ ./pweek3
Parent process started.
Parent PID: 578

--- Parent Process ---
Parent PID: 578
Child PID: 579
Parent is waiting for child...

--- Child Process ---
Child PID: 579
Parent PID (PPID): 578
Child process is running...
Child process terminated.
Parent process terminated.
```

Observation:

The program was executed successfully using the fork() system call. A parent process with PID 1146 created a child process with PID 1147. The child's PPID was 1146, confirming the parent-child relationship. The parent waited while the child executed, and after the child completed its execution, the parent process also terminated.

Result: The experiment successfully demonstrated process creation, PID and PPID identification, parent-child execution, and process termination using fork() and wait().

Task2: Design an experiment Observe process state transitions (Ready, Running, Waiting, Terminated) using ps, top, and /proc, and document the observations.

```

saivihamsh@Varshini-PC:~/OSSP/practical/week3$ cd ~/OSSP/practical/week3
saivihamsh@Varshini-PC:~/OSSP/practical/week3$ mkdir -p task2
saivihamsh@Varshini-PC:~/OSSP/practical/week3$ cd task2
saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$ pwd
/home/saivihamsh/OSSP/practical/week3/task2
saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$ nano process_states.c
saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$ gcc process_states.c -o process_states
saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$ ./process_states &
[1] 1234
Process started.
PID: 1234
PPID: 488
Process is running.
Process will now enter waiting state.
saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$ ps -p 1234 -o pid,ppid,stat,cmd
  PID   PPID  STAT  CMD
   1234   488   S     ./process_states
saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$ grep "^State" /proc/1234/status
State:   S (sleeping)
saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$ top
top - 01:21:50 up 3 min, 1 user, load average: 0.23, 0.25, 0.27
Tasks: 162 total, 1 running, 29 sleeping, 0 stopped, 0 zombie
%Cpu(s): 2.0 us, 1.1 sy, 0.0 ni, 96.5 id, 0.0 wa, 0.1 hi, 0.3 si, 0.0 st
MiB Mem : 3895.6 total, 2417.2 free, 947.7 used, 530.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 2845.8 avail Mem
saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$

```

PID	USER	PR	NI	VIRT	RES
87	root	20	0	35348	12304
1	root	20	0	24192	15388
2	root	20	0	3180	2204
6	root	20	0	3180	2044
49	root	19	-1	50400	16664
81	systemd+	20	0	22416	14488
107	root	20	0	2888	1944
108	root	20	0	4472	3108
109	message+	20	0	8820	5416
115	root	20	0	44096	29880
129	root	20	0	18440	9452
181	syslog	20	0	220548	5904
190	root	20	0	5492	2704
257	_chrony	20	0	23684	11064
265	root	20	0	123460	32548
279	_chrony	20	0	12096	2464
404	root	20	0	3184	1108
407	root	20	0	3200	1248
408	madhuli+	20	0	6268	5556
410	root	20	0	8644	5544
453	madhuli+	20	0	22336	13516
456	madhuli+	20	0	22912	4088
485	madhuli+	20	0	6136	5448

```

saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$ kill 1254
saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$ ps -p 1254
  PID TTY          TIME CMD
saivihamsh@Varshini-PC:~/OSSP/practical/week3/task2$

```

Observation: The process was successfully monitored using ps, top, and /proc. The process was observed in the running and waiting states during execution. The S state indicated that the process was sleeping/waiting. After using the kill command, the process was terminated successfully.

Result: The different process states were successfully observed and verified using Linux process-monitoring commands.